

6 The cell membrane consists of phospholipids and proteins.

(a) Name the current model for the structure of the cell membrane.

(1)

(b) Describe the arrangement of phospholipids in the cell membrane in this model.

(3)

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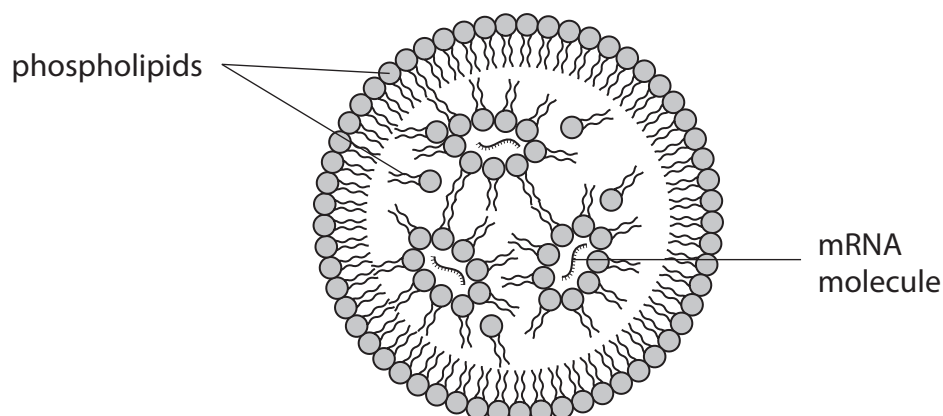
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(c) Phospholipids can be used to form lipid nanoparticles.

Lipid nanoparticles can be used to carry nucleic acids to tissues in the body.

The diagram shows a lipid nanoparticle containing messenger RNA (mRNA) molecules.



- (i) Lipid nanoparticles are very small.

Calculate the diameter of a lipid nanoparticle with a volume of $0.108 \mu\text{m}^3$.

Use the formula

$$V = \frac{4}{3}\pi r^3$$

Use $\pi = 3.0$

(3)

Answer μm

- (ii) Describe the structure of a mRNA molecule.

(3)

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- (iii) Explain the arrangements of phospholipids in this lipid nanoparticle that carries mRNA to tissues in the body.

(3)

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(Total for Question 6 = 13 marks)

